CORE PURE / FURTHER CALCULUS

Integration by Substitution

Mark scheme 15 questions · 107 marks

Questions

Question	Marks by part	Marks	Page
Question 1		6	4
Question 2	(a) 3 (b) 5	8	5
Question 3	(i) 5 (ii) 6	11	6
Question 4		6	7
Question 5	(i) 4 (ii) 6	10	8
Question 6		6	9
Question 7		6	10
Question 8		6	11
Question 9	(a) 3 (b) 5	8	12
Question 10		6	13
Question 11		6	14
Question 12		7	15
Question 13		6	16
Question 14		8	17
Question 15		7	18
	Total	**107**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$u = x + 1 \Rightarrow x - 1 = u - 2,\ \mathrm{d}x = \mathrm{d}u$	**M1**	Uses $u = x + 1$, or another substitution removing the root, expressing $x - 1$ and $\mathrm{d}x$ in terms of u. Condone one slip.
	$\int \frac{(u-2)^2}{\sqrt{u}}\,\mathrm{d}u$	**A1**	Correct integral in u alone, oe. Limits not needed.
	$\int \left(u^{3/2} - 4u^{1/2} + 4u^{-1/2}\right)\mathrm{d}u = \ldots$	**M1**	Writes the integrand as a sum of powers of u and integrates, raising at least one power by 1. Condone expansion slips.
	$\left[\frac{2}{5}u^{5/2} - \frac{8}{3}u^{3/2} + 8u^{1/2}\right]$	**A1**	Correct, oe; need not be simplified.
	$\left(\frac{64}{5} - \frac{64}{3} + 16\right)$ $-\left(\frac{8\sqrt{2}}{5} - \frac{16\sqrt{2}}{3} + 8\sqrt{2}\right)$	**dM1**	Substitutes limits matching their substitution (2 and 4 for $u = x + 1$), or 1 and 3 after reverting to x, and subtracts.
	$= \frac{112}{15} - \frac{64\sqrt{2}}{15} = \frac{16}{15}(7 - 4\sqrt{2})$	**A1 AG**	Exact values at both limits leading without error to the printed result; combined steps accepted. A0 for decimal evaluation.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2(a)	$\mathrm{d}u = 2x\,\mathrm{d}x,\ x^2 = u - 1$ $\int \frac{x^8}{1+x^2}\,x\,\mathrm{d}x = \int \frac{(u-1)^4}{u}\cdot\frac{\mathrm{d}u}{2}$	**M1**	Uses $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$ oe and $x^2 = u - 1$ to write the whole integrand in u. Condone a slip in the factor $\frac{1}{2}$.
	$x = 0 \Rightarrow u = 1,\ x = 2 \Rightarrow u = 5$	**B1**	Both; may be seen only as the limits of the integral.
	$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_1^5 \frac{(u-1)^4}{2u}\,\mathrm{d}u$	**A1 AG**	Correct substitution of every factor, including $\mathrm{d}x$, leading without error to the printed integrand.
		[3]	
2(b)	$\frac{(u-1)^4}{u} = u^3 - 4u^2 + 6u - 4 + \frac{1}{u}$	**M1**	Expands $(u-1)^4$ and divides each term by u, giving five terms including $\frac{1}{u}$. Condone coefficient slips.
	$\frac{1}{2}\left[\frac{u^4}{4} - \frac{4u^3}{3} + 3u^2 - 4u + \ln u\right]$	**M1 A1**	**M1** Integrates term by term, with at least one power increased by 1 and $\frac{1}{u} \to \ln u$. **A1** Correct integral oe; need not be simplified.
	$\left(\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 + \frac{1}{2}\ln 5\right)$ $-\left(\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 + \frac{1}{2}\ln 1\right)$	**dM1**	Substitutes $u = 5$ and $u = 1$ into their integral and subtracts. Not for the limits 0 and 2.
	$= \frac{70}{3} + \frac{1}{2}\ln 5$	**A1**	cso. Exact $A = \frac{70}{3}$ and $B = \frac{1}{2}$, oe.
		[5]	
	Total for Question 2	**8**	

Question	Scheme	Marks	Notes
3(i)	$x^2+4x+8=(x+2)^2+4$	**M1**	Completes the square to the form $(x+2)^2+k$. Condone a slip in k.
	$\int_{-2}^{0}\frac{1}{(x+2)^2+4}\,\mathrm{d}x$ $=\left[\frac{1}{2}\arctan\left(\frac{x+2}{2}\right)\right]_{-2}^{0}$	**M1 A1**	**M1** Integrates their completed square to the form $p\arctan\left(\frac{x+2}{q}\right)$, by formula or substitution.
	$\frac{1}{2}\left(\arctan 1-\arctan 0\right)$	**dM1**	Substitutes both limits into their arctan expression and subtracts, or uses limits changed for a substitution.
	$=\frac{\pi}{8}$	**A1**	cso.
		[5]	
3(ii)	$x=2\sin\theta,\ \mathrm{d}x=2\cos\theta\,\mathrm{d}\theta$ $\int\frac{4\sin^2\theta}{2\cos\theta}\cdot 2\cos\theta\,\mathrm{d}\theta$	**M1**	Uses a substitution such as $x=2\sin\theta$, replacing x, $\sqrt{4-x^2}$ and $\mathrm{d}x$. Condone a coefficient slip in $\mathrm{d}x$.
	$=\int 4\sin^2\theta\,\mathrm{d}\theta$	**A1**	oe, with or without limits.
	$2\int(1-\cos 2\theta)\,\mathrm{d}\theta$	**M1**	Uses $\sin^2\theta=\frac{1}{2}(1-\cos 2\theta)$ and integrates to the form $p\theta+q\sin 2\theta$. Condone sign and coefficient slips.
	$=2\left[\theta-\frac{1}{2}\sin 2\theta\right]$	**A1**	oe.
	$x=0\Rightarrow\theta=0,\ x=\sqrt{3}\Rightarrow\theta=\frac{\pi}{3}$ $2\left(\frac{\pi}{3}-\frac{1}{2}\sin\frac{2\pi}{3}\right)-0$	**dM1**	Uses the changed limits 0 and $\frac{\pi}{3}$ in their integral, or returns to x and uses 0 and $\sqrt{3}$.
	$=\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$	**A1**	cso. Any exact equivalent.
		[6]	
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4	$x = 4\cosh u \Rightarrow \mathrm{d}x = 4\sinh u\,\mathrm{d}u$ $I = \int \frac{4\sinh u}{16\cosh^2 u} \cdot 4\sinh u\,\mathrm{d}u$	**M1**	Substitutes $x = 4\cosh u$ fully, with $\mathrm{d}x = k\sinh u\,\mathrm{d}u$ and $\sqrt{x^2-16}$ as a multiple of $\sinh u$. Condone coefficient slips.
	$= \int \tanh^2 u\,\mathrm{d}u$	**A1**	oe, with no incorrect coefficients seen.
	$= \int \left(1 - \operatorname{sech}^2 u\right)\mathrm{d}u = \ldots$	**M1**	Uses $\tanh^2 u = 1 - \operatorname{sech}^2 u$ and integrates to the form $u \pm k\tanh u$. Condone sign and coefficient slips.
	$= u - \tanh u + c$	**A1**	
	$u = \operatorname{arcosh}\left(\frac{x}{4}\right)$, $\tanh u = \frac{\sqrt{x^2-16}/4}{x/4}$	**M1**	Uses $u = \operatorname{arcosh}\left(\frac{x}{4}\right)$ and $\tanh u$ in terms of x from $\cosh u = \frac{x}{4}$, in their result of the form $u \pm k\tanh u$.
	$I = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c$	**A1 AG**	Correct substituted integral leading without error to the printed result, with $+c$. Condone no justification that $\sinh u > 0$.
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5(i)	$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$ $\Rightarrow \ln(x+1) - \ln(x+2)$	**M1 A1**	**M1** Splits into partial fractions $\frac{A}{x+1} + \frac{B}{x+2}$ and integrates to logarithms. Condone slips in A and B.
	$(\ln 3 - \ln 4) - (\ln 2 - \ln 3)$	**dM1**	Substitutes both limits into their $p\ln(x+1) + q\ln(x+2)$ and subtracts. Collecting the logarithms is not needed.
	$\ln \frac{9}{8}$	**A1**	Or an equivalent with the logarithms collected, such as $2\ln 3 - 3\ln 2$.
		[4]	
5(ii)	$x = 2\sin\theta,\ \mathrm{d}x = 2\cos\theta\,\mathrm{d}\theta$ $\Rightarrow \int \frac{2\cos\theta}{4\sin^2\theta} \cdot 2\cos\theta\,\mathrm{d}\theta$	**M1**	Substitutes $x = 2\sin\theta$ oe throughout, with $\mathrm{d}x = k\cos\theta\,\mathrm{d}\theta$. Condone coefficient slips.
	$\int \cot^2\theta\,\mathrm{d}\theta$	**A1**	Correct integrand in θ, oe such as $\frac{\cos^2\theta}{\sin^2\theta}$.
	$\int(\mathrm{cosec}^2\theta - 1)\,\mathrm{d}\theta = \ldots$	**M1**	Uses $\cot^2\theta = \pm\mathrm{cosec}^2\theta \pm 1$ and integrates to the form $p\cot\theta + q\theta$. Condone sign slips.
	$-\cot\theta - \theta$	**A1**	
	$\theta = \frac{\pi}{6},\ \frac{\pi}{3}$ $\Rightarrow \left(-\frac{1}{\sqrt{3}} - \frac{\pi}{3}\right) - \left(-\sqrt{3} - \frac{\pi}{6}\right)$	**dM1**	Uses the limits $\frac{\pi}{6}$ and $\frac{\pi}{3}$, or returns to x, substituting both into their $p\cot\theta + q\theta$ and subtracting.
	$\frac{2\sqrt{3}}{3} - \frac{\pi}{6}$	**A1**	cso. Or any exact equivalent, such as $\frac{2}{\sqrt{3}} - \frac{\pi}{6}$.
		[6]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6	$x = 4\sinh u \Rightarrow \mathrm{d}x = 4\cosh u\,\mathrm{d}u$ $\sqrt{x^2+16} = 4\cosh u$ $\int \frac{16\sinh^2 u}{4\cosh u}\,4\cosh u\,\mathrm{d}u$	**M1**	Uses $x = 4\sinh u$, or another workable substitution, replacing x, $\sqrt{x^2+16}$ and $\mathrm{d}x$ in terms of u. Condone coefficient slips.
	$= 16\int \sinh^2 u\,\mathrm{d}u$	**A1**	
	$8\int(\cosh 2u - 1)\,\mathrm{d}u$ $= 8\left(\frac{1}{2}\sinh 2u - u\right)$	**M1**	Uses $\sinh^2 u = \pm\frac{1}{2}(\cosh 2u \pm 1)$, or exponentials, and integrates to the form $p\sinh 2u + qu$. Condone sign and coefficient slips.
	$= 4\sinh 2u - 8u + c$	**A1**	oe
	$8\sinh u\cosh u - 8u$ $= 8\left(\frac{x}{4}\right)\left(\frac{\sqrt{x^2+16}}{4}\right) - 8\,\mathrm{arsinh}\left(\frac{x}{4}\right)$	**M1**	Uses $\sinh 2u = 2\sinh u\cosh u$, $\cosh u = \frac{1}{4}\sqrt{x^2+16}$ and $u = \mathrm{arsinh}\,\frac{x}{4}$ on their result of the form $p\sinh 2u + qu$.
	$= \frac{1}{2}x\sqrt{x^2+16} - 8\,\mathrm{arsinh}\left(\frac{x}{4}\right) + c$	**A1 AG**	Printed result, including $+c$, reached without error from a correct integral in u; combined steps accepted.
	Total for Question 6	**6**	

Question	Scheme	Marks	Notes
7	$u = e^x \Rightarrow \mathrm{d}x = \frac{\mathrm{d}u}{u}$ $\int \frac{u^3+2u^2}{\sqrt{u+4}} \cdot \frac{1}{u}\,\mathrm{d}u$	**M1**	Replaces the exponentials by powers of u and $\mathrm{d}x$ by $\frac{\mathrm{d}u}{u}$ oe. Condone slips in powers. Also for $u = e^x + 4$ directly.
	$= \int \frac{u^2+2u}{\sqrt{u+4}}\,\mathrm{d}u$	**A1**	oe, limits not required. After $v = e^x + 4$ directly, $\frac{(v-4)^2+2(v-4)}{\sqrt{v}}$ oe.
	$v = u + 4 \Rightarrow \int \frac{(v-4)^2+2(v-4)}{\sqrt{v}}\,\mathrm{d}v$ $= \int \left(v^{3/2} - 6v^{1/2} + 8v^{-1/2}\right)\mathrm{d}v$ $= \ldots$	**M1**	Writes the integrand as a sum of powers of v and integrates, raising at least one power by 1. Condone expansion slips.
	$\left[\frac{2}{5}v^{5/2} - 4v^{3/2} + 16v^{1/2}\right]$	**A1**	oe, limits not required.
	$\left(\frac{2}{5}\cdot 6^{5/2} - 4\cdot 6^{3/2} + 16\cdot 6^{1/2}\right)$ $-\left(\frac{2}{5}\cdot 5^{5/2} - 4\cdot 5^{3/2} + 16\cdot 5^{1/2}\right)$	**dM1**	Substitutes limits 5 and 6 in v, or the original limits after substituting back, into their integral and subtracts.
	$= \frac{32}{5}\sqrt{6} - 6\sqrt{5} = \frac{32\sqrt{6}-30\sqrt{5}}{5}$	**A1 AG**	Correct substituted expression leading without error to the printed result in exact surds; combined steps accepted. A decimal check alone earns nothing.
	Total for Question 7	**6**	

Question	Scheme	Marks	Notes
8	$\mathrm{d}u = \frac{2}{3}\operatorname{sech}^2 x\,\mathrm{d}x,\ \tanh x = \frac{3}{2}u$ $\Rightarrow \int \frac{\frac{3}{2}}{\sqrt{9-4\left(\frac{3}{2}u\right)^2}}\,\mathrm{d}u$	**M1**	Replaces $\operatorname{sech}^2 x\,\mathrm{d}x$ by $k\,\mathrm{d}u$ and $\tanh x$ by a multiple of u, giving an integral in u only. Condone slips in the constants.
	$= \frac{1}{2}\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u$	**A1**	Correct integrand in u, oe.
	$= \frac{1}{2}\arcsin u$	**M1 A1**	**M1** Integrates their integral of the form $\frac{p}{\sqrt{a^2-b^2u^2}}$ to $\lambda\arcsin(\mu u)$.
	$x = 0 \Rightarrow u = 0,\ x = 1 \Rightarrow u = \frac{2}{3}\tanh 1$ $\frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right) - \frac{1}{2}\arcsin 0$	**dM1**	Applies u-limits 0 and $\frac{2}{3}\tanh 1$ to their arcsin form, or substitutes back for u and applies 0 and 1.
	$= \frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right)\ [\approx 0.266]$	**A1**	oe, or awrt 0.266.
	Total for Question 8	**6**	

Question	Scheme	Marks	Notes
9(a)	$x = \frac{u^2-1}{2},\ \mathrm{d}x = u\,\mathrm{d}u$ $\int \frac{\frac{1}{2}(u^2-1)}{2+u}\,u\,\mathrm{d}u$	**M1**	Replaces x, $\sqrt{2x+1}$ and $\mathrm{d}x$ by expressions in u, with $\mathrm{d}x$ of the form $ku\,\mathrm{d}u$.
	$x = 0 \Rightarrow u = 1,\ x = \frac{15}{2} \Rightarrow u = 4$	**B1**	
	$\int_1^4 \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u$	**A1 AG**	Correct substituted integral leading without error to the printed integrand.
		[3]	
9(b)	$u^3 - u = (u+2)(au^2+bu+c) + d \Rightarrow \ldots$	**M1**	Divides $u^3 - u$ by $u+2$ by any method, reaching a quadratic quotient and a constant remainder. Condone slips.
	$\frac{u(u^2-1)}{2(u+2)} = \frac{1}{2}\left(u^2 - 2u + 3 - \frac{6}{u+2}\right)$	**A1**	Or quotient $u^2 - 2u + 3$ and remainder -6 stated.
	$\frac{1}{2}\left[\frac{u^3}{3} - u^2 + 3u - 6\ln(u+2)\right]_1^4$	**M1**	Integrates their quotient and $\frac{d}{u+2}$, with $\frac{d}{u+2} \to d\ln(u+2)$ and at least one power increased by 1. Condone coefficient slips.
	$\frac{1}{2}\left[\left(\frac{52}{3} - 6\ln 6\right) - \left(\frac{7}{3} - 6\ln 3\right)\right]$ $\ln 6 - \ln 3 = \ln 2 \Rightarrow \ldots$	**dM1**	Substitutes 4 and 1 into their integral, subtracts, and combines their logarithms into a single logarithm.
	$\frac{15}{2} - 3\ln 2$	**A1**	cso. Or $A = \frac{15}{2}$, $B = 3$. Not for $\frac{15}{2} - \ln 8$ or a decimal.
		[5]	
	Total for Question 9	**8**	

Question	Scheme	Marks	Notes
10	$u = 1 + \sin x \Rightarrow \mathrm{d}u = \cos x \, \mathrm{d}x$	**M1**	Uses an appropriate substitution such as $u = 1 + \sin x$, replacing $\cos x \, \mathrm{d}x$ by a multiple of $\mathrm{d}u$.
	$\cos 2x = 1 - 2(u-1)^2$ $I = \int \frac{-2u^2+4u-1}{u^{3/2}} \, \mathrm{d}u$	**A1**	Correct integrand wholly in u, oe. Need not be expanded.
	$\int \left(-2u^{1/2} + 4u^{-1/2} - u^{-3/2}\right) \mathrm{d}u$	**M1**	Divides their quadratic in u by $u^{3/2}$ term by term and integrates, raising the power of at least one term by 1.
	$= \left[-\frac{4}{3}u^{3/2} + 8u^{1/2} + 2u^{-1/2}\right]$	**A1**	oe, need not be simplified.
	$\left(-\frac{4}{3}\left(\frac{3}{2}\right)^{3/2} + 8\left(\frac{3}{2}\right)^{1/2} + 2\left(\frac{3}{2}\right)^{-1/2}\right)$ $-\left(-\frac{4}{3} + 8 + 2\right)$	**dM1**	Applies limits 1 and $\frac{3}{2}$ to their integral, or 0 and $\frac{\pi}{6}$ after substituting back, subtracting the right way round.
	$= -\frac{26}{3} + \frac{11}{3}\sqrt{6}$	**A1**	cso. In the form $a + b\sqrt{6}$, or $a = -\frac{26}{3}$, $b = \frac{11}{3}$ stated; $\frac{11\sqrt{6}-26}{3}$ alone scores A0.
	Total for Question 10	**6**	

Question	Scheme	Marks	Notes
11	$x = 4\sin\theta \Rightarrow \mathrm{d}x = 4\cos\theta\,\mathrm{d}\theta$ $\int \frac{16\sin^2\theta}{\sqrt{16-16\sin^2\theta}}\,4\cos\theta\,\mathrm{d}\theta$	**M1**	Uses $x = 4\sin\theta$ or $x = 4\cos\theta$, replacing x^2, the root and $\mathrm{d}x$. Condone coefficient slips.
	$16\int \sin^2\theta\,\mathrm{d}\theta$	**A1**	oe, such as $16\cos^2\theta$ from $x = 4\cos\theta$.
	$8\int (1-\cos 2\theta)\,\mathrm{d}\theta = \ldots$	**M1**	Uses $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and integrates to the form $p\theta + q\sin 2\theta$. Condone sign and coefficient slips.
	$8\left[\theta - \frac{1}{2}\sin 2\theta\right]$	**A1**	oe, such as $8\theta - 4\sin 2\theta$.
	$x = 0 \Rightarrow \theta = 0$, $x = 2\sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$ $8\left(\frac{\pi}{4} - \frac{1}{2}\sin\frac{\pi}{2}\right) - 8(0-0)$	**dM1**	Converts both limits to θ, or returns to x, and subtracts. Not for the x-limits put into a function of θ.
	$2\pi - 4$	**A1**	cso. Accept $-4 + 2\pi$, or $a = -4$, $b = 2$.
	Total for Question 11	**6**	

Question	Scheme	Marks	Notes
12	$u = \frac{3}{2}\tanh x$, $\mathrm{d}u = \frac{3}{2}\operatorname{sech}^2 x\,\mathrm{d}x$ $\Rightarrow \int \frac{\frac{2}{3}}{\sqrt{4-9\left(\frac{2u}{3}\right)^2}}\,\mathrm{d}u$	**M1**	Substitutes for $\tanh x$, such as $u = k\tanh x$, replacing both $\operatorname{sech}^2 x\,\mathrm{d}x$ and $\tanh^2 x$ in the new variable. Condone coefficient slips.
	$= \frac{1}{3}\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u$	**A1**	Correct integrand for their substitution, oe such as $\frac{1}{\sqrt{4-9u^2}}$ for $u = \tanh x$.
	$e^{2x} = 2+\sqrt{3}$ $\Rightarrow \tanh x = \frac{1+\sqrt{3}}{3+\sqrt{3}} = \frac{1}{\sqrt{3}}$ $\Rightarrow u = \frac{\sqrt{3}}{2}$	**B1**	Exact upper limit for their variable, from exponentials; $\frac{1}{\sqrt{3}}$ for $u = \tanh x$. Not from a calculator decimal alone.
	$\frac{1}{3}\arcsin u$	**M1 A1**	**M1** Integrates their $\frac{p}{\sqrt{a^2-b^2u^2}}$ to $k\arcsin(\lambda u)$. Condone coefficient slips. **A1** oe in their variable, such as $\frac{1}{3}\arcsin\frac{3u}{2}$ for $u = \tanh x$.
	$\frac{1}{3}\left(\arcsin\text{“}\frac{\sqrt{3}}{2}\text{”} - \arcsin 0\right)$	**dM1**	Substitutes both limits, in the variable of their integral, into their arcsin form and subtracts.
	$\frac{\pi}{9}$	**A1**	
	Total for Question 12	**7**	

Question	Scheme	Marks	Notes
13	$x = 2\sec\theta \Rightarrow \mathrm{d}x = 2\sec\theta\tan\theta\,\mathrm{d}\theta$ $\int \frac{\sqrt{4\sec^2\theta - 4}}{4\sec^2\theta}\,2\sec\theta\tan\theta\,\mathrm{d}\theta$	**M1**	Replaces x, $\mathrm{d}x$ and $\sqrt{x^2-4}$, with $\mathrm{d}x = k\sec\theta\tan\theta\,\mathrm{d}\theta$. Condone a slip in simplifying the root.
	$= \int \frac{\tan^2\theta}{\sec\theta}\,\mathrm{d}\theta$	**A1**	oe, such as $\frac{\sin^2\theta}{\cos\theta}$.
	$\int(\sec\theta - \cos\theta)\,\mathrm{d}\theta = \ldots$	**M1**	Uses $\tan^2\theta = \sec^2\theta - 1$ oe and integrates both terms, $\sec\theta$ to any form of its standard integral. Condone sign and coefficient slips.
	$\ln\lvert\sec\theta + \tan\theta\rvert - \sin\theta$	**A1**	oe
	$\theta = 0,\ \frac{\pi}{3}$ $\left(\ln(2+\sqrt{3}) - \sin\frac{\pi}{3}\right) - (\ln 1 - \sin 0)$	**dM1**	Applies limits 0 and $\frac{\pi}{3}$ to their integral and subtracts, or returns to x and uses 2 and 4.
	$\ln(2+\sqrt{3}) - \frac{\sqrt{3}}{2}$	**A1**	cso. Accept simplified exact equivalents such as $-\ln(2-\sqrt{3}) - \frac{\sqrt{3}}{2}$.
	Alternative: Substitution $x = 2\cosh u$		
	$x = 2\cosh u \Rightarrow \mathrm{d}x = 2\sinh u\,\mathrm{d}u$ $\int \frac{\sqrt{4\cosh^2 u - 4}}{4\cosh^2 u}\,2\sinh u\,\mathrm{d}u$	**M1**	Replaces x, $\mathrm{d}x$ and $\sqrt{x^2-4}$, with $\mathrm{d}x = k\sinh u\,\mathrm{d}u$. Condone a slip in simplifying the root.
	$= \int \tanh^2 u\,\mathrm{d}u$	**A1**	oe, such as $\frac{\sinh^2 u}{\cosh^2 u}$.
	$\int(1 - \operatorname{sech}^2 u)\,\mathrm{d}u = \ldots$	**M1**	Uses $\tanh^2 u = 1 - \operatorname{sech}^2 u$ oe and integrates to the form $pu + q\tanh u$. Condone sign and coefficient slips.
	$u - \tanh u$	**A1**	
	$u = 0,\ \operatorname{arcosh} 2 = \ln(2+\sqrt{3})$ $\ln(2+\sqrt{3}) - \tanh(\operatorname{arcosh} 2) - 0$	**dM1**	Applies limits 0 and arcosh 2 to their integral and subtracts, or returns to x and uses 2 and 4.
	$\ln(2+\sqrt{3}) - \frac{\sqrt{3}}{2}$	**A1**	cso. Accept simplified exact equivalents such as $-\ln(2-\sqrt{3}) - \frac{\sqrt{3}}{2}$.
	Total for Question 13	**6**	

Question	Scheme	Marks	Notes
14	$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ $= \sin\theta\left(3 - 4(1-\cos^2\theta)\right)$	**M1**	Uses a triple-angle or compound-angle expansion and $\sin^2\theta = 1-\cos^2\theta$ to write $\sin 3\theta$ as $\sin\theta$ times a function of $\cos\theta$. Condone sign slips.
	$\sin 3\theta = \sin\theta(4\cos^2\theta - 1)$	**A1**	
	$u = \cos\theta \Rightarrow \mathrm{d}u = -\sin\theta\,\mathrm{d}\theta$ $\theta = 0 \Rightarrow u = 1$, $\theta = \frac{\pi}{2} \Rightarrow u = 0$	**M1**	Uses $u = \cos\theta$ with $\mathrm{d}u = \pm\sin\theta\,\mathrm{d}\theta$ to reach an integral in u alone, and changes the limits or later returns to θ.
	$\int_0^1 \frac{4u^2-1}{u+2}\,\mathrm{d}u$	**A1**	oe such as $-\int_1^0$; the sign from $\mathrm{d}u$ and the order of the limits must be consistent.
	$\frac{4u^2-1}{u+2} = 4u - 8 + \frac{15}{u+2}$	**M1**	Divides by $u+2$, or substitutes $v = u+2$, and integrates, including a term $k\ln(u+2)$. Condone coefficient slips.
	$\left[2u^2 - 8u + 15\ln(u+2)\right]_0^1$	**A1**	
	$(2 - 8 + 15\ln 3) - 15\ln 2$	**dM1**	Substitutes both limits into their integral and subtracts.
	$= 15\ln\left(\frac{3}{2}\right) - 6$	**A1 AG**	Correct substituted expression leading without error to the printed result, logarithms combined.
	Total for Question 14	**8**	

Question	Scheme	Marks	Notes
15	$u = \sin\theta \Rightarrow \mathrm{d}u = \cos\theta\,\mathrm{d}\theta$ $\cos^5\theta = (1-u^2)^2\cos\theta$	**M1**	Uses $u = \sin\theta$ with $\mathrm{d}u = \cos\theta\,\mathrm{d}\theta$ and $\cos^2\theta = 1-u^2$ to reach an integrand in u only. Condone sign slips.
	$\int_0^1 \frac{(1-u^2)^2}{2+u}\,\mathrm{d}u$	**A1**	Correct integral with limits 0 and 1, oe; numerator may be expanded.
	$\frac{u^4-2u^2+1}{u+2} = \ldots + \frac{\ldots}{u+2}$	**M1**	Divides their quartic by $u+2$, by any method, to a cubic quotient and a constant remainder. Condone slips.
	$u^3 - 2u^2 + 2u - 4 + \frac{9}{u+2}$	**A1**	
	$\left[\frac{u^4}{4} - \frac{2u^3}{3} + u^2 - 4u + 9\ln(u+2)\right]_0^1$	**M1**	Integrates their quotient, raising at least one power by 1, and their remainder $\frac{k}{u+2}$ to $k\ln(u+2)$.
	$\left(\frac{1}{4} - \frac{2}{3} + 1 - 4 + 9\ln 3\right) - 9\ln 2$	**dM1**	Substitutes $u=1$ and $u=0$, or $\theta = \frac{\pi}{2}$ and 0 after reversing the substitution, and subtracts.
	$= 9\ln\left(\frac{3}{2}\right) - \frac{41}{12}$	**A1 AG**	cso. Correct substituted expression, including $9\ln 2$ from the lower limit, leading without error to the printed result.
	Total for Question 15	**7**	